

Date : Ex :	Import and Export in Python as CSV,XLSX AND SQL
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AIM :

To perform data import and export operations using Python and Pandas by transferring the Instagram Analytics dataset between CSV, Excel, and SQL databases, and to verify the imported data through basic data analysis.

ALGORITHM :

1. Start the Python/Jupyter Notebook environment.
2. Import the required libraries such as pandas and sqlite3.
3. Load the Instagram Analytics dataset into a Pandas DataFrame.
4. Export the DataFrame to an Excel file using `to_excel()`.
5. Export the DataFrame to a CSV file using `to_csv()`.
6. Create/connect to an SQLite database using `sqlite3`.
7. Export the DataFrame into an SQL table using `to_sql()`.
8. Import the data from the SQL table into a Pandas DataFrame using `read_sql()`.
9. Import the exported CSV file using `read_csv()`.
10. Display the first few records using `head()`.
11. Display dataset information using `info()`.
12. Generate basic statistical information using `describe()`.
13. Export the SQL-imported data back to CSV/Excel for verification.
14. Close the database connection.
15. Stop.

PROGRAM :

```
[4]: #IMPORT CSV FILE
import pandas as pd
df = pd.read_csv("Instagram_Analytics.csv")
```

```

: #EXPORT CLSX FILE
df.to_excel("Instagram_Analytics_Sample.xlsx", index=False)
print(df)
print(df.head())
print(df.info())
print(df.describe())
print("Excel file exported successfully!")

```

mean	8466.952498	0.042107	502.152805	120.081169
std	6825.900355	0.024095	290.364038	11.009108
min	327.000000	0.000000	0.000000	70.000000
25%	4102.000000	0.024000	251.000000	113.000000
50%	6595.000000	0.040600	501.000000	120.000000
75%	10620.500000	0.057000	755.000000	127.000000
max	105203.000000	0.271000	1000.000000	166.000000

```

      hashtags_count
count  29999.000000
mean      7.987933
std       2.828130
min       0.000000
25%       6.000000
50%       8.000000
75%      10.000000
max      21.000000
Excel file exported successfully!

```

```

: #READ EXCEL
df = pd.read_excel("Instagram_Analytics_Sample.xlsx")

df.head()

```

	post_id	account_id	account_type	follower_count	media_type	content_category	traffic_source	has_cal
0	IG0000001	7	brand	3551	reel	Technology	Home Feed	
1	IG0000002	20	creator	31095	image	Fitness	Hashtags	
2	IG0000003	15	brand	8167	reel	Beauty	Reels Feed	
3	IG0000004	11	creator	9044	carousel	Music	External	
4	IG0000005	8	creator	15986	reel	Technology	Profile	

5 rows x 23 columns

```
import sqlite3

# Connect to SQLite database
conn = sqlite3.connect("Instagram_Analytics.db")

# Export DataFrame to SQL table
df.to_sql("instagram_analytics", conn, if_exists="replace", index=False)

print("Data exported successfully to SQL!")

conn.close()
```

Data exported successfully to SQL!

```
: import sqlite3
import pandas as pd

# Connect to database
conn = sqlite3.connect("Instagram_Analytics.db")

# Import SQL table into DataFrame
df_sql = pd.read_sql("SELECT * FROM instagram_analytics", conn)

print("Data imported successfully from SQL!")
print(df_sql.head())

conn.close()
```

Data imported successfully from SQL!

	post_id	account_id	account_type	follower_count	media_type	\
0	IG0000001	7	brand	3551	reel	
1	IG0000002	20	creator	31095	image	
2	IG0000003	15	brand	8167	reel	
3	IG0000004	11	creator	9044	carousel	
4	IG0000005	8	creator	15986	reel	

	content_category	traffic_source	has_call_to_action	post_datetime	\
0	Technology	Home Feed	1	2024-11-30 06:00:00	
1	Fitness	Hashtags	1	2025-08-15 15:00:00	
2	Beauty	Reels Feed	0	2025-09-11 16:00:00	
3	Music	External	0	2025-09-18 03:00:00	
4	Technology	Profile	0	2025-03-21 09:00:00	

	post_date	...	comments	shares	saves	reach	impressions	\
0	2024-11-30	...	5	7	34	4327	6230	
1	2025-08-15	...	10	21	68	7451	8268	
2	2025-09-11	...	2	1	22	1639	2616	
3	2025-09-18	...	0	7	0	2877	3171	
4	2025-03-21	...	8	5	21	5350	8503	

```
#EXPORT CSV
```

```
df.to_csv("Instagram_Analytics_Sample.csv", index=False)
```

```
print("CSV file exported successfully!")
```

CSV file exported successfully!

```
#IMPORT CSV
```

```
import pandas as pd
```

```
df_csv = pd.read_csv("Instagram_Analytics_Sample.csv")
```

```
print("CSV file imported successfully!")
```

```
print(df_csv.head())
```

```
print(df_csv.info())
```

```
print(df_csv.describe())
```

CSV file imported successfully!

	post_id	account_id	account_type	follower_count	media_type	\
0	IG0000001	7	brand	3551	reel	
1	IG0000002	20	creator	31095	image	
2	IG0000003	15	brand	8167	reel	
3	IG0000004	11	creator	9044	carousel	
4	IG0000005	8	creator	15986	reel	

	content_category	traffic_source	has_call_to_action	post_datetime	\
0	Technology	Home Feed	1	2024-11-30 06:00:00	
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3	Music	External	0	2025-09-18 03:00:00	
4	Technology	Profile	0	2025-03-21 09:00:00	

	post_date	...	comments	shares	saves	reach	impressions	\
0	2024-11-30	...	5	7	34	4327	6230	
1	2025-08-15	...	10	21	68	7451	8268	

RESULT :

The Instagram Analytics dataset was successfully imported and exported between Pandas, CSV, Excel, and SQL. The imported data was verified using `head()`, `info()`, and `describe()`, confirming that the data was successfully transferred and retained its structure.